

HARDIK GOEL

Data & AI Architecture Leader | Enterprise AI Platforms, Agentic Systems & MLOps Governance | 12+ Years

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PROFESSIONAL SUMMARY

Data and AI architecture leader with 12+ years building enterprise data, ML and GenAI platforms, and the teams and governance that run them. At Tesco, own end-to-end architecture of an AI-driven commodity forecasting platform supporting **£3.5B** in annual procurement, delivered from the Bengaluru GCC to UK commercial stakeholders. At Paytm, built the incentives and merchant data capability from zero to a **14-member** engineering team, **promoted twice** while owning the charter, hiring bar, roadmap and delivery: payment SLA cut from **15 days to 3** and platform cost reduced by **over 50%**. Designed one of the earliest enterprise implementations of a persona-based agentic engineering framework. In parallel, author and deliver enterprise AI capability programmes, including architect-level curricula that prepare senior architects for Forward Deployed Engineer roles. Chief Technology Officer Programme, ISB. Published author.

CORE COMPETENCIES

Leadership & Strategy	Technology Strategy & Roadmapping, Engineering Management, Greenfield Capability Build (0 to 14), Role Sequencing & Technical Hiring, Distributed / GCC Delivery Model, Executive Stakeholder Management, Architecture Ownership, Engineering Governance, Enterprise AI Capability Building, Mentoring
Architecture	Solution Architecture, Enterprise Architecture, Data Architecture, Platform Strategy, Reference Architecture, Reusable Framework Design, Engineering Standards, Scalable Systems, HLD/LLD, Data Contracts, Schema Evolution
AI & GenAI	Agentic AI, RAG, LLMs (GPT, Claude, Gemini), MCP, Vector Databases, Embeddings, Multimodal Agents, Conversational AI, AI Orchestration, AI-Augmented Engineering
AI Governance	Responsible AI, EU AI Act, ISO 42001, DPDP Act, Explainability (SHAP), Audit Lineage (Microsoft Purview), Maker-Checker Controls
ML & Modelling	Feature Engineering & Selection, Overfitting Control (Time-Series Cross-Validation, Regularisation), Evaluation Metrics (MAPE/WMAPE, Precision-Recall, AUC), Baseline Benchmarking, Drift Detection (PSI), Champion-Challenger, LightGBM, scikit-learn
MLOps	MLflow, PSI Drift Monitoring, Model Registry, Feature Stores, Azure ML, DQ Gates, Champion-Challenger Deployment
Data Engineering	Apache Spark, Kafka, Airflow, DBT, Databricks, Apache Hudi, Flink, NiFi, Sqoop, Delta Lake, Data Mesh, Data Lakehouse, CDC, Streaming
Cloud & Infra	AWS (EMR, Lambda, Glue, Athena, Redshift, S3, RDS, OpenSearch, ECS), Azure, Databricks, Docker, Kubernetes, CI/CD, GitOps
Languages	Python (PySpark, Pandas, NumPy, scikit-learn), Java (Spring Boot), SQL, Scala

PROFESSIONAL EXPERIENCE

TESCO | Bengaluru, India | Technical Architect, Data & AI

Jan 2025 - Present

*End-to-end architecture and modelling ownership of an AI-driven commodity forecasting platform supporting a **£3.5B** retail procurement domain, built in Tesco's Bengaluru GCC for commercial and procurement stakeholders based in the UK.*

Prescriptive Decision Loop: Architecture (End-to-End Ownership)

- Architected and delivered a production-grade AI commodity forecasting platform supporting **£3.5B** in annual procurement across multiple commodities and prediction horizons, designed as a closed prescriptive decision loop rather than a forecast dashboard: ingestion and feature engineering, multi-horizon forecast generation, buy-timing and coverage recommendations surfaced to commercial buyers, and realised-outcome capture fed back into monitoring and retraining.
- Owned every layer of that loop end to end: reference architecture, data contracts and schema evolution, feature pipelines, model training and promotion, serving and consumption, and the governance wrapper around all of it, with no separate platform, ML or DevOps team to hand off to.
- Built reusable metadata-driven data and ML pipelines integrating LSEG and enterprise RDBMS sources, cutting commodity onboarding effort by **65%** and improving scalability across new forecasting use cases.
- Revamped fragmented forecasting workflows into a scalable, production-ready architecture optimised for reliability, maintainability, auditability and developer velocity.
- Delivered stakeholder analytics dashboards and data enrichment solutions to UK-based commercial teams, improving commercial visibility into commodity trends and model outputs.

Modelling Depth & MLOps Governance

- Owned the modelling lifecycle hands-on, not just its orchestration: feature engineering and selection across commodity price drivers and market indicators, multicollinearity and leakage checks on lagged and rolling features, and deliberate feature-set pruning where added inputs bought accuracy on paper but not out of sample.
- Controlled overfitting through time-series cross-validation with out-of-time holdouts, regularisation and complexity tuning, validating every candidate model against naive and seasonal baselines using MAPE/WMAPE before it could be called an improvement.
- Instrumented interpretability and decay: SHAP-based attribution to explain individual buy recommendations to commercial stakeholders, and PSI drift monitoring on both input and prediction distributions, wired to champion-challenger promotion so a challenger replaced an incumbent only on evidence.
- Designed governed MLOps workflows on Azure ML with MLflow: DQ gates, model metadata registries, security scanning, audit logs, Microsoft Purview lineage and Maker-Checker deployment controls.
- Defined and enforced engineering governance standards across Data Engineering and ML (modular architecture, typed interfaces, reusable pipeline templates and automated pytest coverage gates), cutting deployment failures by **80%**.

Agentic Framework Design & AI-Augmented Engineering

- Designed a persona-based agentic framework with specialised AI roles for Data Engineering, ML Review, Security and MLOps Governance, one of the earliest enterprise implementations of role-based AI agents in a production data platform.
- Pioneered AI-augmented engineering workflows using Claude and GPT agents for automated code review, test generation, documentation and pipeline scaffolding, lifting delivery velocity by **40%**.
- Authored AI-enforceable markdown standards enabling automated enforcement of coding practices, security guardrails and quality gates through structured context orchestration and RAG workflows.
- Contributed architecture consulting and platform engineering across enterprise PoCs and production adoption initiatives, accelerating AI adoption across cross-functional teams.
- Designing and delivering "AI Architecture 101", an AI literacy programme for up to **1,000 ERP users** across five batches (Oct to Nov 2026).

PAYTM | Bengaluru & Noida, India

Jan 2022 - Jan 2025

Data Engineering Manager

Jul 2024 - Jan 2025

Senior Technical Data Engineering Lead

Apr 2023 - Jul 2024

Technical Data Engineering Lead

Jan 2022 - Apr 2023

*Brought in to stand up the data and analytics capability for the incentives and merchant charter from a standing start. Owned the charter, architecture, hiring sequence and delivery, and scaled it into a **14-member** engineering team in Bengaluru serving product, finance and business stakeholders at Noida HQ.*

Greenfield Capability Build & Distributed Team Ownership

- Stood up a greenfield analytics and data capability from zero: defined the charter and success metrics, wrote the reference architecture and first-year roadmap, and delivered a working thin slice into production before scaling headcount, so hiring was justified by a proven path, not by a plan.
- Sequenced roles deliberately rather than hiring a full-stack org on day one. Hired senior data engineers first to make ingestion, orchestration and data contracts trustworthy; added an analytics and serving-layer engineer once there was governed data worth exposing; brought in ML and rule-engine specialists only when a modelled decision was blocking business value; and deferred dedicated BI, QA and specialist analyst roles until volume, not optics, justified the hire.
- Set the technical bar personally and kept it: designed and ran live SQL, Python/PySpark and system-design interview rounds across all levels, watching candidates write and debug against a real problem to see how they reasoned rather than testing recall.
- Ran the distributed model explicitly, with engineering in Bengaluru and stakeholders in Noida: fixed overlap hours, written decision records and design docs as the default over meetings, a demo-and-signoff cadence tied to sprint boundaries, and a single named escalation path, so remote stakeholders never lost context.
- Organised the team into outcome-owning pods with clear ownership boundaries, on-call rotation and Maker-Checker governance; mentored engineers into senior and lead roles and grew internal successors for platform ownership.
- Stayed hands-on throughout as a working manager: reviewing PySpark and SQL in code review, debugging production pipeline failures alongside the team, and personally owning the hardest architectural decisions.

Incentive Management Platform

- Architected and delivered a high-throughput incentive processing platform: reduced payment SLA from **15 days to 3**, improved accuracy to **over 97%**, and cut attrition by **40%**.
- Designed complex rule-engine pipelines using Apache Airflow, AWS EMR, PySpark and MongoDB; used DBT for SQL transformations and conducted the Databricks DBR impact assessment.
- Drove **over 50%** cost reduction on the Incentives platform; architected PGOLAP on Databricks DBR, cutting costs by **over 70%** for **110M daily events** at Dream11 scale (1.8B events/day).

- Built FSEBuddy, a self-service platform enabling Product team autonomy and reducing engineering dependency for routine data tasks.
- Delivered CustomerEchoEase, a GenAI-powered customer insights solution, as an internal innovation initiative.
- Implemented observability frameworks, Maker-Checker processes and process-level governance controls across the platform.

Merchant Analytics Platform

- Architected a real-time merchant analytics and cross-sell recommendation platform, driving **over 40%** offline merchant sales growth and **over 30%** increase in cross-sell rate.
- Engineered high-throughput event processing using Kafka, NiFi, Spark, Apache Hudi and Elasticsearch to power sub-second API queries for merchant dashboards.

IMPETUS TECHNOLOGIES | Gurgaon, India | Senior Software Engineer

Feb 2020 - Jan 2022

US Airline: Data Lake Governance Framework (ICON)

- Architected ICON (Ingest, Catalog, Optimize, Notify), a data lake governance framework reducing downstream ingestion TAT by **over 40%** and ingestion error rates by **over 90%**.
- Implemented governance controls using AWS Lambda, Glue, Athena, RDS and PySpark, with automated access controls and quality scans for all ingested datasets.
- Led agile delivery, sprint planning and stakeholder requirements gathering across US-based client teams.

Intelligence & Investigation: Search Engine Optimisation

- Delivered a **99% latency reduction** (from 2 seconds to 20 ms) on a sensitive-data search engine using Apache Solr and the ELK stack.
- Architected storage tiering: searchable parameters in Solr/ELK, less-searchable parameters in Apache Cassandra, with Kafka as the data backbone.

Semiconductor Client: Real-Time Sensor Analytics

- Built real-time sensor data ingestion and anomaly detection dashboards using Kafka, Apache Flink and QuickSight; warehoused data via Redshift Spectrum with DBT transformations; orchestrated via MWAA and Apache Beam.
- Developed an internal NLP sentiment analyser using NLTK, outperforming VADER accuracy benchmarks.

COGNIZANT TECHNOLOGY SOLUTIONS | Gurgaon, India | Associate, Big Data Engineer

Apr 2018 - Jan 2020

Anti-Money Laundering (AML): European Bank

- Reduced the **91%** AML false-positive rate, enabling faster regulatory reporting and cutting the monthly investigation backlog against a 30-day SLA target.
- Implemented an ML pipeline using LightGBM (Paylay model) with data ingestion from Oracle, Teradata and sandpits via Sqoop, Oozie, Hive/Impala and Apache Spark pre-processing.

ACCENTURE SOLUTIONS | Pune & Gurgaon, India | Software Engineering Analyst

Oct 2014 - Mar 2018

Promoted from Associate Software Engineer in Apr 2016.

- Built CDR/UDR ingestion pipelines for a US telecom client (postpaid) using Apache Kafka, Spark and Hive.
- Led cross-functional sprint, system and UAT testing programmes across multiple US and UK telecom clients using Agile/Scrum.
- Delivered MVNO mobile automation frameworks using Selenium WebDriver, Appium, TestNG and Jenkins.

TOLEXO ONLINE PVT. LTD. | Noida, India | Associate Software Programmer

Apr 2014 - Sep 2014

- Started my career at a B2B e-commerce startup, a subsidiary of IndiaMART InterMESH.

ADDITIONAL EXPERIENCE (PARALLEL TO EMPLOYMENT)

Enterprise AI Capability Programmes | Independent Practice

2015 - Present

- Design and deliver enterprise AI and Data programmes for engineering and leadership audiences across pharma, healthcare, retail, consulting and IT services organisations; around **200 delivered hours** in Aug and Sep 2026 alone.
- Authored a **9-week Level 1 AI Architect programme** (18 sessions, 60% hands-on) that graduates senior architects into Forward Deployed Engineer roles through a formal Architect Review Board; designed the five-dimension scoring rubric and a Kirkpatrick Level 1 to 5 measurement model.
- Authored a 9-week AI Solution Architect cohort programme, delivered live to two parallel batches (**108 delivered hours**), with six tracked competencies, weekly graded work products and peer-review scorecards.
- Delivered a **72-hour** advanced architecture and AI-assisted engineering intensive covering C4 modelling, ADRs, GenAI architecture patterns, evaluation harnesses, Responsible AI (ISO 42001, EU AI Act), AI threat modelling (OWASP Top 10 for LLM Applications) and spec-driven development.
- Earlier curricula and cohorts across UpGrad, Analytics Vidhya, GUVI, Interview Kickstart, Ducat, DataCouch and ToolsQA, including live coding evaluations and mock technical interviews; PAN-India corporate workshops and guest lectures at graduate universities.

Forward-Deployed Client Engagements | Independent

- Beverage distribution business: took requirements first-hand from client leadership, then architected and delivered a GST-compliant distributor management platform, now live in production: **4,150+** tax invoices and **18,000+** stock movements processed with full audit traceability across **686 outlets** and **INR 1.3 crore** in invoiced sales, replacing manual spreadsheet and WhatsApp order tracking.
- Multi-property hospitality group: designing and building a hospitality management system across five properties, working directly with the owners on priorities and rollout.

Author & Speaker

- Author, Forgotten By Design: The Silent Epidemic of Digital Dementia (CleverFox Publishing, Oct 2025), available on Amazon India and US.
- Keynote panelist, Elastic CXO Fusion (GenAI track); podcast guest; technical writer for the Times of India and ToolsQA.
- Weekly writing on enterprise AI at goelh.substack.com and medium.com/@hardik.goel214.

PROJECTS

CompliSense AI | complisenseai.com

AI governance and compliance platform for regulated industries.

- Agentic compliance monitoring, alerting and remediation against the EU AI Act and India's DPDP Act, for finance, retail and healthcare.

Switchboard: AI Model Routing & Orchestration Framework | github.com/hardik-goel

Open-source intelligent routing framework for provider abstraction, dynamic execution strategies and AI workflow optimisation.

- Provider-agnostic architecture routing across OpenAI, Anthropic, Google and other LLM providers.
- Dynamic execution strategies including fallback chains, cost-optimised routing and latency-aware model selection, designed for enterprise AI platform integration.

http-queryable | github.com/hardik-goel

Open-source support for the HTTP QUERY method in Node.

- Node implementation of the HTTP QUERY method (RFC 10008) with correct body-aware caching; adapters for Express, Fastify and raw http plus an isomorphic client.

AudioSummariser: GenAI Audio Intelligence Framework | github.com/hardik-goel | PyPI

Production-grade audio intelligence pipeline, published as a PyPI package.

- End-to-end audio-to-insight pipeline: transcription, embedding, chunking, summarisation and conversational RAG retrieval, with vector database retrieval and LLM-powered sentiment analysis.

CommodityBuddy: AI-Powered Procurement Intelligence

GenAI assistant for commodity forecasting and procurement insights, bridging enterprise data systems with conversational AI.

- Conversational AI layer over commodity forecasting outputs, enabling non-technical stakeholders to query model insights in natural language (structured data, embeddings, RAG, conversational interface).

Agentic AI Platform

Multi-agent orchestration for enterprise knowledge work.

- Multi-agent orchestration framework with specialised AI personas for enterprise knowledge work, with MCP-based integrations and autonomous workflow execution.

EDUCATION

Chief Technology Officer Programme

Sep 2026 - Apr 2027

Indian School of Business (ISB) | Technology Leadership and Strategy

Bachelor of Technology, Computer Science & Engineering

2010 - 2014

Krishna Engineering College (Uttar Pradesh Technical University)

CERTIFICATIONS

Astronomer Certification: DAG Authoring for Apache Airflow 3	Oct 2025
Amazon Q Developer Fundamentals	Oct 2025
Astronomer Certification for Apache Airflow 3	Jul 2025
Databricks Generative AI Fundamentals	Jul 2023
Databricks Spark Structured Streaming	Jul 2023
Databricks Certified Associate Developer for Apache Spark 3.0	Jul 2022
Microsoft Certified: Azure Fundamentals (AZ-900)	Aug 2020